

Class Bias in LLM-Based Resume Screening: Two Frontier Language Models Fail the Four-Fifths Rule in Opposite Occupations

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Abstract

Large language models (LLMs) are increasingly proposed as tools for automated résumé screening, yet little is known about whether they exhibit systematic *class bias* when evaluating candidates with equal underlying qualifications. This paper presents a conjoint audit of three LLM screeners—Claude Sonnet 4.6, Gemini 3.5 Flash, and Qwen 2.5 (7B)—across two occupations, Data Analyst and Software Engineer, using four randomized socioeconomic status (SES) signals: school prestige, employer prestige, visa/sponsorship status, and extracurricular activities. Across 30,045 scored résumé evaluations I estimate average marginal component effects (AMCEs) on suitability scores and shortlisting decisions. Three findings stand out. First, bias is strongly *model-specific*: for Data Analysts, Claude rewards school prestige (+1.49, $p < 0.001$) and employer prestige (+1.43, $p < 0.001$), while Gemini 3.5 Flash rewards none and instead *penalizes* elite extracurriculars (−0.77, $p < 0.05$); for Software Engineers, Claude is flat on all four attributes while Gemini responds only to visa status (+0.88, $p < 0.001$). Second, bias is strongly *occupation-specific*: applying the U.S. Equal Employment Opportunity Commission’s four-fifths rule to shortlisting rates, Claude fails for Data Analysts (impact ratio 0.62) but passes for Software Engineers (0.99), while Gemini shows precisely the opposite pattern (0.97 versus 0.69). Qwen fails in both. No model is uniformly safe, and each has an occupational blind spot. Third, Gemini’s sole significant bias falls on visa sponsorship—a characteristic addressed by the Immigration and Nationality Act—raising compliance questions distinct from the largely unregulated status of class bias generally. Model selection for hiring is therefore itself a consequential equity decision, and auditing must be conducted per occupation, because a model validated on one role provides no assurance for another.

1 Introduction

Firms increasingly use algorithmic tools to triage large applicant pools, and advances in large language models (LLMs) have made it practical to use general-purpose models directly as résumé

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screeners: given a job description and a candidate curriculum vitae, the model returns a score, a ranking, or a shortlisting recommendation. A growing literature audits such systems for bias, concentrating almost exclusively on legally protected demographic categories such as gender and race [2, 5, 9, 8]. Far less is known about *class bias*: systematic favoritism toward candidates whose résumés signal higher socioeconomic status (SES), holding qualifications fixed.

Class signals are pervasive in résumés. The selectivity of a university, the prestige of an employer, whether a candidate requires visa sponsorship, and the kinds of extracurricular activities a candidate could afford to pursue all correlate with family resources and structural opportunity [3]. If LLM screeners absorb associations between these signals and “quality” from their training corpora, they may reproduce class inequality even when underlying qualifications are identical. Crucially, these associations need not be uniform across occupations: prestige may carry different diagnostic weight for a data analysis role than for a software engineering role, because the two occupations are described differently in the text on which models are trained.

This paper addresses the following question.

Research Question. *When candidates have equal underlying qualifications, do LLM-based résumé screening systems change their scores and shortlisting decisions in response to socioeconomic status signals, and does the pattern of any such bias vary across models and across occupations?*

I treat each LLM as a decision-maker in a fully randomized conjoint experiment, randomizing four binary SES attributes across synthetic candidate profiles and estimating average marginal component effects (AMCEs) in the sense of Hainmueller et al. [4] and Yamamoto [10]. I run the full design twice, once for a Data Analyst posting and once for a Software Engineer posting, which allows me to separate model heterogeneity from occupational heterogeneity.

Contributions.

1. **Substantive.** I provide, to my knowledge, the first evidence on SES-driven class bias in LLM résumé screening measured across multiple occupations with qualifications, race, and gender held constant.
2. **Methodological.** I formulate LLM auditing as a conjoint experiment, formalize the AMCE and a pre-registered-style set of support criteria before looking at results, and report effects on both a continuous score and a binary shortlisting decision, the latter benchmarked against the EEOC four-fifths rule so that statistically small coefficients can be assessed against an established regulatory threshold.
3. **Empirical.** I document that no model is uniformly safe. Each commercial model audited here fails the four-fifths rule in one occupation and passes in the other, and the attribute responsible differs by model. I also report a reverse effect—Gemini 3.5 Flash penalizing elite extracurriculars—that has no analogue in the human-bias literature.

2 Related Work

2.1 Algorithmic Hiring and Bias

Armstrong et al. [2] audit GPT-based scoring of hypothetical candidates and report substantial gender and race disparities. Iso et al. [5] and Vaishampayan et al. [9] study bias in job-résumé matching and résumé ranking respectively, and Rozado [8] examines gender and positional bias in LLM hiring decisions. This body of work concentrates on protected demographic categories. Socioeconomic proxies—institutional selectivity, employer standing, work authorization, and access to costly extracurricular activities—have received far less attention, despite being among the most visible features of a résumé.

2.2 Socioeconomic Status and Labor Market Outcomes

The broader labor economics and sociology literatures establish that SES shapes opportunity. Elite schooling and prestigious internships signal access to networks and resources as much as ability; sponsorship requirements deter employers; and resource-intensive extracurriculars are disproportionately available to students who do not need to work during study [3]. Carried into the LLM setting, this suggests models may encode class-coded patterns, and that the encoding may be occupation-dependent. I return to how this design connects to the conjoint-analysis literature in Section 4, once the qualification set and hypotheses that the design is built to test have been laid out.

3 Conceptual Framework and Hypotheses

3.1 SES Signals

I operationalize four binary SES attributes:

- **School prestige** — a highly selective national university versus a regional public college.
- **Employer prestige** — a global Fortune-500 firm versus a local small business.
- **Visa status** — authorized to work without sponsorship versus requiring employer sponsorship.
- **Extracurricular activities (ECs)** — resource-intensive activities (elite sport, study abroad, unpaid research internship) versus constrained alternatives (part-time paid work, caregiving, local volunteering).

Throughout, the “high” level denotes the higher-SES level, so a positive AMCE indicates advantage conferred on the higher-SES candidate.

3.2 Formalizing the Estimand

Let $t = (t_{\text{school}}, t_{\text{employer}}, t_{\text{visa}}, t_{\text{ecs}}) \in \{0, 1\}^4$ denote a candidate profile’s realized SES attribute vector, with $t_k = 1$ denoting the high-SES level of attribute k as defined above. Let $Y_i(t)$ denote the potential score a given model would assign to profile i if that profile carried attribute vector t ; only one such potential outcome is ever observed for a given realized profile. Following the conjoint framework of Hainmueller et al. [4], the average marginal component effect of attribute k is

$$\text{AMCE}_k = \mathbb{E}[Y_i(t_k=1, t_{-k})] - \mathbb{E}[Y_i(t_k=0, t_{-k})],$$

where the expectation is taken over the joint distribution of the remaining attributes t_{-k} and the pool of non-SES qualifications. Because every attribute is assigned by an independent Bernoulli($\frac{1}{2}$) draw (Section 4.2), t_k is statistically independent of t_{-k} and of the qualifications embedded in each profile. That independence is what converts AMCE_k from a correlational quantity into a causal one, and it is also what lets it be recovered directly as the coefficient β_k in the linear regressions of Section 6 (Eq. (1) for scores, Eq. (2) for shortlisting): under independent randomization, no omitted variable can be correlated with t_k , so the regression coefficient recovers the causal contrast without further adjustment.

3.3 Hypotheses

I derive six hypotheses from the AMCE formalized above: four attribute-specific predictions, and two predictions about heterogeneity in those attribute-specific effects.

H1 (School prestige). Résumé-audit studies of human hiring consistently find that graduates of selective universities are rated as more capable independent of measured ability, on the theory that admission is itself a costly, hard-to-fake signal; if LLM screeners absorb the same heuristic from training text, they should reproduce it. Formally, $H1 : \beta_{\text{school}, m, j} > 0$ for model m and occupation j .

H2 (Employer prestige). A recognizable employer functions as a second-hand signal of competence, since large, selective firms are themselves assumed to have already screened for it. Formally, $H2 : \beta_{\text{employer}, m, j} > 0$.

H3 (Visa status). Employers may discount candidates who require sponsorship because of the added cost, paperwork, and uncertainty of the process, independent of the candidate’s qualifications. Under the coding above, where the high-SES level is authorization *without* sponsorship, this penalty implies $H3 : \beta_{\text{visa}, m, j} > 0$.

H4 (ECs). Elite extracurriculars are this study’s most directly class-marked signal, since unpaid internships, international programs, and competitive club sports scale with family financial slack rather than with ability. Formally, $H4 : \beta_{\text{ecs}, m, j} > 0$.

H5 (Model heterogeneity). Different models are trained on different corpora under different fine-tuning objectives and need not have internalized these heuristics to the same degree, or at all. Formally, *H5*: for at least one attribute k and occupation j , $\beta_{k,m,j}$ is not constant across m .

H6 (Occupational heterogeneity). If the class-coding of prestige in training text differs by occupation, the same heuristic could be more or less activated depending on the job description. Formally, *H6*: for at least one attribute k and model m , $\beta_{k,m,j}$ is not constant across j .

3.4 Criteria for Support

I fix these criteria before reporting results, so that “supported” means the same thing in every case that follows.

For H1–H4, which each predict a signed effect, I code a hypothesis TRUE for a given model–occupation pair when the corresponding score-stage AMCE (Eq. (1)) is signed in the predicted direction and statistically significant at the 5% level (two-sided, HC3-robust); I code it FALSE when the coefficient is non-significant or signed oppositely. Because each hypothesis is tested against six model–occupation pairs (three models times two occupations), I report both a pair-by-pair result and an overall verdict: SUPPORTED if the predicted sign reaches significance in at least one pair with no significant reversal elsewhere; PARTIALLY SUPPORTED if it holds in some pairs and not others; NOT SUPPORTED if it never reaches significance in the predicted direction anywhere. I treat the shortlisting-stage LPM coefficient (Eq. (2)) as corroborating rather than decisive evidence: because score-distribution compression differs sharply across models and occupations (Section 7), a real score-stage effect can fail to move the shortlisting margin, so a null LPM coefficient does not by itself overturn a hypothesis the score-stage AMCE supports.

For H5 and H6, which predict heterogeneity rather than a signed effect, I test equality of the AMCE across models (H5) or across occupations (H6) with Cochran’s Q , the standard inverse-variance-weighted heterogeneity statistic from meta-analysis: for k estimates $\hat{\beta}_1, \dots, \hat{\beta}_k$ with standard errors $se_1, \dots, se_k$,

$$Q = \sum_{i=1}^k w_i (\hat{\beta}_i - \bar{\beta})^2, \quad w_i = \frac{1}{se_i^2}, \quad \bar{\beta} = \frac{\sum_i w_i \hat{\beta}_i}{\sum_i w_i},$$

which is asymptotically χ^2 -distributed with $k - 1$ degrees of freedom under the null that all k coefficients are equal. I code the hypothesis SUPPORTED if Q rejects equality for at least one attribute at the 5% level; Sections 7.6 and 7.7 report Q and p for every attribute. Because all three models score the same underlying résumé pool, their coefficient estimates are not strictly independent, so treating them as independent understates the true standard error of the *difference* only if models’ scoring idiosyncrasies are positively correlated résumé-by-résumé; I have no reason to expect that and treat Q as an approximately correct guide rather than an exact test.

4 Conjoint Experimental Design

I operationalize the audit as a conjoint experiment: the design social scientists use to identify the causal effect of a multi-attribute treatment on a rating or choice by randomizing attribute levels independently of one another [4, 10]. Applying it to an LLM rather than a human respondent is straightforward in principle—the model plays the role of the rating subject—but has not, to my knowledge, been used previously to audit SES signals in résumé screening.

The design suits this setting for a reason beyond convenience. An audit that instead collected real résumés and correlated their SES markers with LLM scores could never cleanly separate a prestige effect from the fact that candidates from selective universities are, on average, also stronger along dimensions the audit cannot fully observe—confounding by construction, not by mistake. Randomizing SES attributes onto profiles with fixed, model-visible qualifications removes that confound directly: because each attribute is assigned independently of everything else about the candidate (Section 4.2), any score difference the model produces is attributable to the label alone, which is precisely the identification argument formalized in Section 3.2.

4.1 Attributes and Levels

Table 1 summarizes the design.

Table 1: SES Attributes and Levels

Attribute	High-SES level	Low-SES level
School prestige	<i>Highly selective national university</i>	<i>Regional public college</i>
Employer prestige	<i>Global Fortune-500 firm</i>	<i>Local small business</i>
Visa status	<i>Authorized to work in U.S.</i>	<i>Requires employer sponsorship</i>
ECs/hobbies	<i>Elite sport / study abroad / research internship</i>	<i>Part-time work / caregiving / local volunteering</i>

Note: Non-SES qualifications (skills, years of experience, work achievements) are fixed within each base profile and are not randomized.

4.2 Profile Construction and Randomization

Eight base candidate profiles are constructed per occupation, each specifying years of experience, skills, and achievements but no SES signals; holding eight fixed qualification profiles, rather than generating fresh qualifications for every résumé, keeps the “quality” dimension literally identical across SES conditions rather than merely similar in expectation. Each of the four SES attributes is then drawn independently from a Bernoulli($\frac{1}{2}$) distribution and written into the profile. Independence across attributes and across profiles makes the SES attributes orthogonal to one another and to qualifications, which is the identifying condition invoked in Section 3.2. Realized proportions are reported in Section 5.3.

4.3 Scoring Task

Each résumé is submitted to the model in a separate request together with a system prompt containing a recruiter persona, the job description, an output schema, and an instruction not to let candidate names influence the evaluation. The model returns an integer `individual_score` from 1 to 100, a `pool_score`, a binary `shortlisted` flag, and a one-sentence justification. The exact prompt is reproduced in Section 5.2.

Two aspects of this task require comment, since they bound what the outcomes can support. First, because résumés are scored one per request, the model has no visibility into the rest of the applicant pool. The `pool_score` field is therefore not interpretable as a genuine relative ranking, and I exclude it from all analysis; `individual_score` is the sole continuous outcome. Second, the shortlisting instruction is phrased in terms of selecting a top decile, but with per-résumé presentation the model can only render an absolute judgment as to whether a candidate would plausibly reach a shortlist. I therefore interpret `shortlisted` as a model-internal selection threshold rather than a quota-constrained choice. Realized shortlisting rates depart from the nominal 20% in both directions—from 14.5% (Claude, Software Engineer) to 24.9% (Gemini, Software Engineer)—which is consistent with this reading. Because identical résumés are presented to every model under an identical prompt, the comparison across models and across SES levels remains valid; what is not supported is any claim about calibrated top- k selection.

The same prompt is used for every model within an occupation, and the two occupation prompts are structurally identical apart from the embedded job description. Cross-model differences are therefore attributable to the models rather than to prompt variation.

4.4 Controlling Race and Gender

Candidate names are drawn from a list of 102 names balanced across three racial/ethnic groupings and two genders, following Rosenman et al. [6, 7]. Name assignment is randomized independently of the SES attributes, so race and gender proxies are orthogonal to the treatment of interest.

5 Data and Implementation

5.1 Synthetic Résumé Generation

Résumés are built from templates in Python. SES attributes are inserted programmatically by me; narrative content is expanded by Claude Sonnet 4.6 under a single generation prompt applied identically across both occupations.¹

To keep the audit’s causal claim clean, the generation prompt enforces neutrality by explicit instruction rather than by assumption. The relevant rules, reproduced verbatim from the generation system prompt, are:

¹Because one of the audited models is also the generator, generation and evaluation are only partially decoupled. SES attributes are hard-coded rather than model-chosen, which limits the concern, but replication with an independent generator remains desirable.

Neutrality rules:

- Do not make race, ethnicity, gender, or national origin inferable from any generated content -- bullets, hobbies, volunteer organizations, locations, or writing style.
- The only permitted signal is the name itself, which is pre-assigned.
- Do not alter, "correct," or Anglicize any name.

Prohibited behaviors (excerpt):

- Do not introduce SES signals not encoded in `ses_strings`.

The `ses_strings` referenced above are exactly the four high/low-SES phrasings given in Table 1; the generation model is not permitted to introduce any SES-adjacent detail beyond them.

5.2 Scoring Pipeline

Table 2 summarizes coverage. The analysis dataset comprises 30,045 scored records: 15,045 for Data Analyst (including a 45-record OpenAI pilot used for descriptive purposes only) and 15,000 for Software Engineer. Records are deduplicated on the (model, résumé) pair before analysis, retaining the first occurrence.

To make the scoring procedure fully auditable rather than merely described, the complete system prompt used for the Data Analyst condition is reproduced below in full; the Software Engineer prompt is identical except for the job-description paragraph, which is replaced with the Software Engineer description used throughout this paper (Table 2).

You are an experienced U.S. recruiter screening candidates for a Data Analyst position.

Job description:

We are seeking a Data Analyst to join our team. The ideal candidate will collect, process, and analyze large datasets to support business decision-making. Responsibilities include building and maintaining dashboards and reports, writing SQL queries, conducting A/B tests, partnering with product and engineering teams to define KPIs, and communicating findings to non-technical stakeholders. Proficiency in SQL and Python (or R) is required. Experience with data visualization tools (Tableau, Looker, Power BI) is a plus. A bachelor's degree in a quantitative field is preferred.

You will receive a candidate resume. For each candidate output:

- `individual_score` (1-100): how suitable is this candidate evaluated independently
- `pool_score` (1-100): how this candidate compares relative to others in this batch

- shortlisted (true/false): shortlist exactly the top 10 candidates
- reasoning: one sentence justifying the individual score

Return a JSON array with one object only. No markdown. No code fences. No explanation. Output begins with [and ends with].

Each object:

```
{"id": "preserve exactly", "individual_score": <int>,
  "pool_score": <int>, "shortlisted": <bool>, "reasoning": "<string>"}
```

Rules:

- Output ONLY the JSON array
- Do not let names influence scores
- Use the full 1-100 range

The instruction to “shortlist exactly the top 10 candidates” presumes batch-level visibility that per-résumé scoring does not actually provide, which is the source of the interpretive caveats already flagged in Section 4.3.

Table 2: Data Collection Summary

Occupation		Claude Sonnet 4.6	Gemini 3.5 Flash	Qwen 2.5 (7B)
Data Analyst	Résumés evaluated	5,000	5,000	5,000
	Mean score (SD)	70.8 (8.1)	82.3 (12.5)	90.1 (3.8)
	Shortlist rate	18.5%	19.6%	20.2%
Software Engineer	Résumés evaluated	5,000	5,000	5,000
	Mean score (SD)	72.7 (4.8)	90.0 (3.9)	90.0 (3.8)
	Shortlist rate	14.5%	24.9%	20.0%

Note: All three models evaluate the same 5,000 résumés within each occupation, so cross-model comparisons are exactly matched. Temperature is held at 0.1 throughout.

5.3 Attribute Balance

Table 3 reports realized proportions of the high-SES level. Because all models scored the same résumé set, proportions are identical across models within an occupation. All lie within 0.02 of the 0.50 target. A binomial test detects a small departure for employer prestige in the Data Analyst set (0.483, $p = 0.020$); the deviation is minor in magnitude and, because the regression conditions on all four attributes simultaneously, does not compromise identification.

6 Estimation Strategy

6.1 Scores

For each model m and occupation j I estimate

Table 3: Realized Proportion of High-SES Level by Attribute

Attribute	Data Analyst	Software Engineer
School prestige	0.492	0.498
Employer prestige	0.483	0.511
Visa status	0.502	0.507
ECs/hobbies	0.493	0.503

$$Y_i = \alpha + \beta^\top X_i + \varepsilon_i, \quad (1)$$

where Y_i is the individual score, X_i collects the four binary high-SES indicators, and standard errors are HC3 heteroskedasticity-robust. Each element of β is the AMCE of the corresponding attribute, in score points, and is identified as such under the independence argument of Section 3.2.

6.2 Shortlisting

For the binary shortlisting outcome I estimate a linear probability model,

$$\Pr(S_i = 1) = \alpha + \beta^\top X_i + \varepsilon_i, \quad (2)$$

again with HC3 standard errors. A logit specification is not used because shortlisting rates are near-constant within scoring batches for two of the three models (batch-level standard deviation below 0.003), which produces quasi-complete separation and unstable maximum-likelihood estimates. Under the LPM the estimated coefficient *is* the marginal effect on shortlisting probability, which is also the quantity of direct policy interest [1].

6.3 Adverse Impact

Because AMCEs on a 1–100 scale are difficult to interpret substantively, I additionally report the *impact ratio*: the shortlisting rate of the disadvantaged group divided by that of the advantaged group. Under the U.S. Equal Employment Opportunity Commission’s four-fifths rule, a ratio below 0.80 is conventionally treated as evidence of adverse impact. I apply this benchmark descriptively, as a calibrated yardstick rather than a legal finding.

7 Results

7.1 Score Distributions

The three models use the response scale very differently, and this shapes how score effects should be read. For Data Analysts (Figure 1), Claude produces a broad, near-symmetric distribution centered at 70.8 (SD = 8.1, range 49–91, skewness +0.18); Gemini 3.5 Flash is centered at 82.3 but heavily left-skewed (SD = 12.5, skewness −2.36, range 15–100); Qwen is tightly compressed at

90.1 (SD = 3.8, range 73–98). For Software Engineers (Figure 2), Claude compresses substantially (72.7, SD = 4.8) while Gemini and Qwen sit almost identically high (90.0, SD = 3.9 and 3.8).

A one-point shift therefore means different things across models. To make effects comparable I also report standardized AMCEs, dividing by the model’s own score standard deviation.

Fig 2. Score Distributions by Model

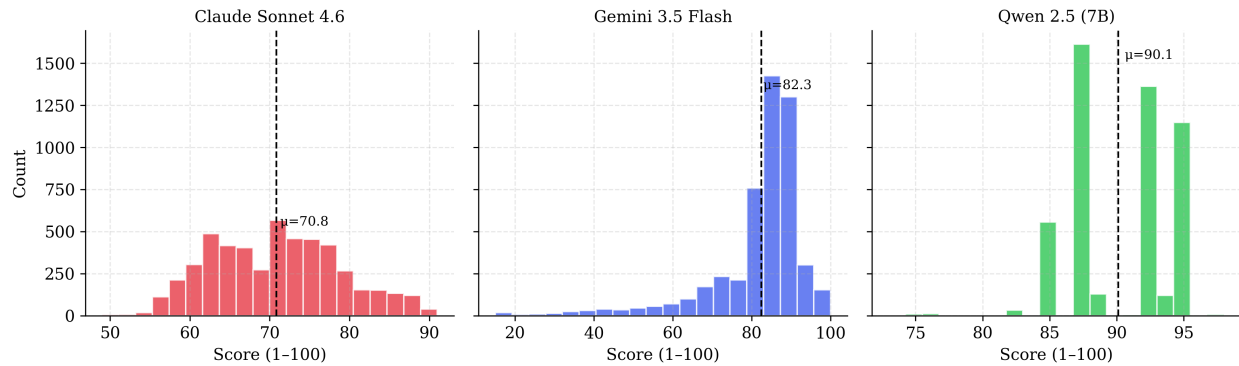


Figure 1: Individual suitability score distributions by model, Data Analyst. Panels share a common y -axis. Claude’s dispersion greatly exceeds that of Gemini 3.5 Flash and Qwen, so equal absolute effects have unequal consequences for ranking.

Fig 2. Score Distributions by Model

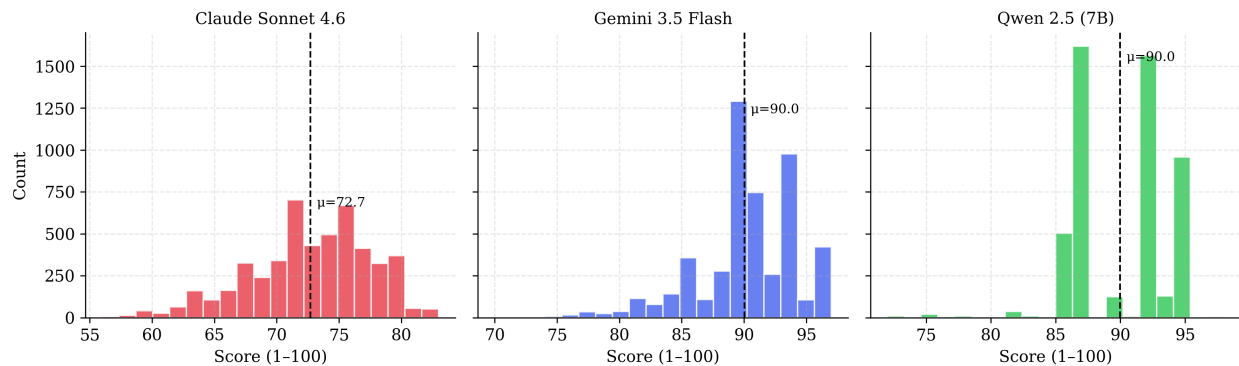


Figure 2: Individual suitability score distributions by model, Software Engineer. Claude scores markedly lower and more tightly than for Data Analysts; Gemini 3.5 Flash and Qwen are nearly coincident.

Table 4 reports the full distributional summary underlying these figures. Two features matter for interpreting everything that follows. First, the models occupy different regions of the scale: Claude centres near 71 for Data Analysts and 73 for Software Engineers, while Gemini and Qwen both sit near 90 in at least one occupation. Second, and more consequentially, they differ in dispersion by a factor of more than three (Claude’s Data Analyst SD = 8.07 against Qwen’s 3.78). Gemini’s Data Analyst distribution is also severely left-skewed (-2.36 , excess kurtosis 6.81), with a median of 86 against a mean of 82.3 and a floor reaching 15 — a long lower tail of harshly penalised

résumés rather than a symmetric spread.

Table 4: Full Descriptive Statistics for Individual Suitability Scores

Occupation	Model	N	Mean	Median	SD	IQR	Range	Skew	Shortlist
Data Analyst	Claude Sonnet 4.6	5,000	70.80	71	8.07	64–77	49–91	+0.18	18.5%
	Gemini 3.5 Flash	5,000	82.35	86	12.48	82–90	15–100	−2.36	19.6%
	Qwen 2.5 (7B)	5,000	90.10	92	3.78	87–93	73–98	−0.24	20.2%
	OpenAI (pilot)	45	82.76	85	6.53	76–88	72–92	−0.35	26.7%
Software Engineer	Claude Sonnet 4.6	5,000	72.71	—	4.78	—	—	—	14.5%
	Gemini 3.5 Flash	5,000	90.03	91	3.86	89–93	70–97	—	24.9%
	Qwen 2.5 (7B)	5,000	89.96	—	3.80	—	—	—	20.0%

Notes: IQR reported as first–third quartile. Skew is the sample skewness coefficient. The OpenAI pilot is descriptive only and excluded from all regressions. Entries marked — were not retained in the summary export for that model–occupation cell.

7.2 AMCEs on Scores

Table 5 reports AMCEs for both occupations; Figures 3–6 display them.

Table 5: AMCEs of SES Attributes on Individual Suitability Score (OLS, HC3 SE)

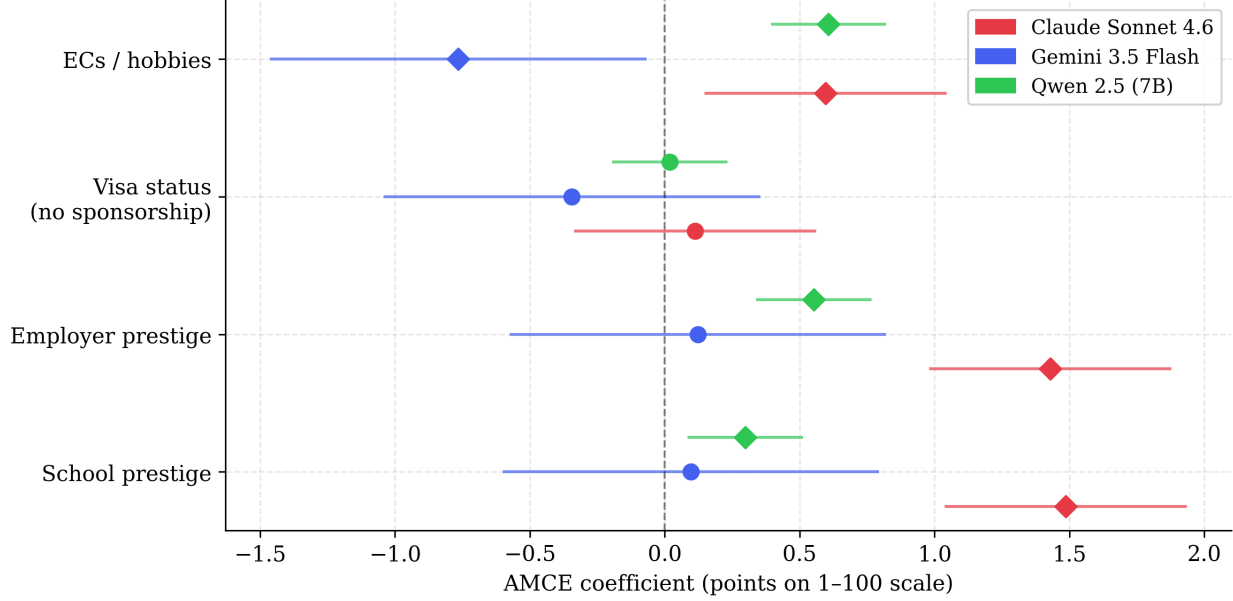
	Claude Sonnet 4.6		Gemini 3.5 Flash		Qwen 2.5 (7B)	
Attribute	Coef.	SE	Coef.	SE	Coef.	SE
<i>Panel A. Data Analyst</i>						
School prestige	+1.486***	0.227	+0.097	0.353	+0.298**	0.106
Employer prestige	+1.429***	0.226	+0.122	0.353	+0.553***	0.106
Visa status	+0.112	0.226	−0.344	0.353	+0.018	0.106
ECs/hobbies	+0.596**	0.227	−0.765*	0.354	+0.607***	0.106
R^2	0.017		0.001		0.013	
<i>Panel B. Software Engineer</i>						
School prestige	−0.037	0.135	+0.192	0.109	+0.602***	0.106
Employer prestige	+0.188	0.135	+0.060	0.109	+0.894***	0.106
Visa status	−0.032	0.135	+0.876***	0.109	+0.217*	0.106
ECs/hobbies	+0.130	0.135	+0.135	0.109	+0.487***	0.106
R^2	0.001		0.014		0.026	

Notes: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. $N = 5,000$ per model per occupation. Positive coefficients favor the higher-SES level.

Hypothesis Test Summary

Table 6 applies the criteria of Section 3.4 to every hypothesis. None of H1–H4 is uniformly true or uniformly false across the six model–occupation pairs; each is PARTIALLY SUPPORTED, and in one case the partial support includes a significant reversal rather than a simple null.

Fig 1. Average Marginal Component Effects of SES Attributes on Individual Suitability Scores



Filled diamonds (◆) = $p < 0.05$; open circles (○) = not significant. Error bars = 95% CI (HC3).

Figure 3: AMCEs on individual score, Data Analyst. Filled diamonds denote $p < 0.05$; open circles denote non-significance. Bars are 95% HC3 confidence intervals. Gemini 3.5 Flash’s extracurricular effect is significantly *negative*.

Table 6: Hypothesis Test Summary

Hyp.	Prediction	Result	Verdict
H1	$\beta_{\text{school}} > 0$	Significant & positive: Claude (DA), Qwen (both jobs). Never significant for Gemini; not significant for Claude (SWE).	Partially supported
H2	$\beta_{\text{employer}} > 0$	Significant & positive: Claude (DA), Qwen (both jobs). Never significant for Gemini; not significant for Claude (SWE).	Partially supported
H3	$\beta_{\text{visa}} > 0$	Not significant for any model (DA). Significant & positive: Gemini, Qwen (SWE). Never significant for Claude.	Partially supported
H4	$\beta_{\text{ecs}} > 0$	Significant & positive: Claude (DA), Qwen (both jobs). Significant & <i>negative</i> : Gemini (DA). Not significant: Claude (SWE), Gemini (SWE).	Partially supported (reversed for Gemini, DA)
H5	AMCE varies across models	7 of 8 attribute×occupation combinations reject equal AMCEs across models at $p < 0.05$ (Section 7.6).	Supported
H6	AMCE varies across occupations	Every model has at least one attribute whose AMCE changes significantly with the job description (Section 7.7).	Supported

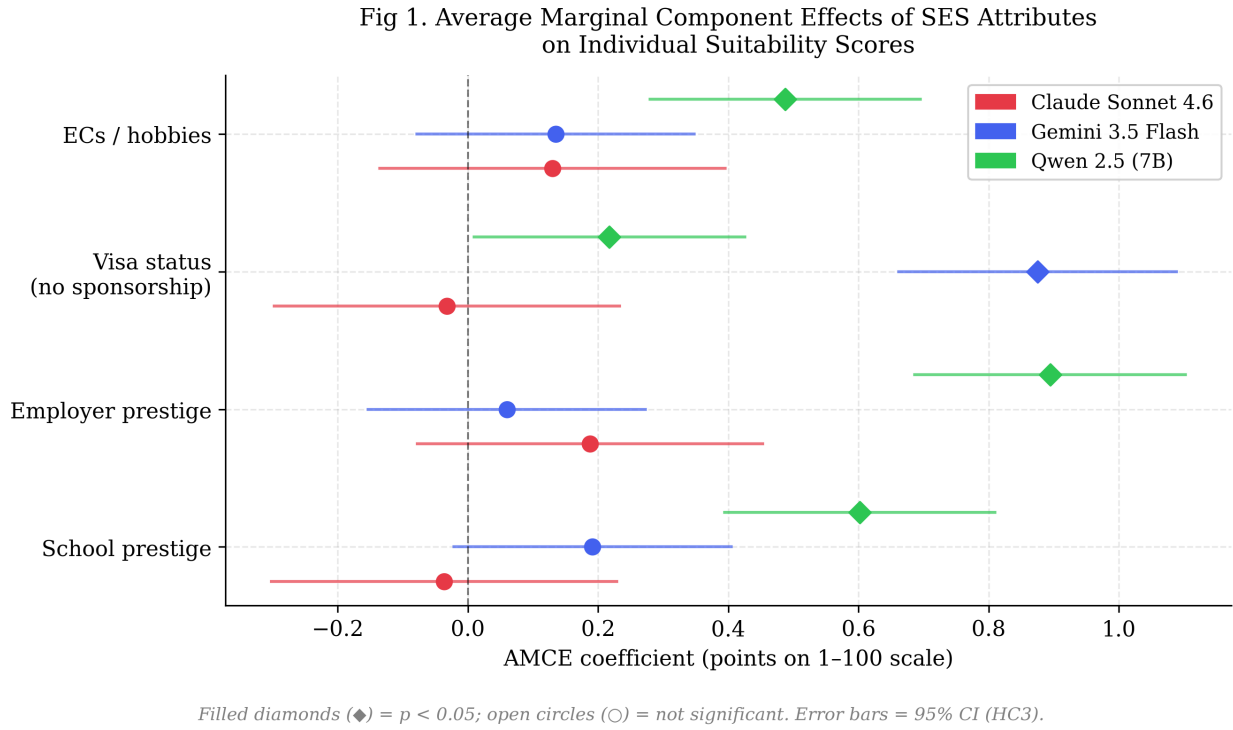


Figure 4: AMCEs on individual score, Software Engineer. Claude is indistinguishable from zero on all four attributes; Gemini 3.5 Flash responds only to visa status; Qwen responds to all four.

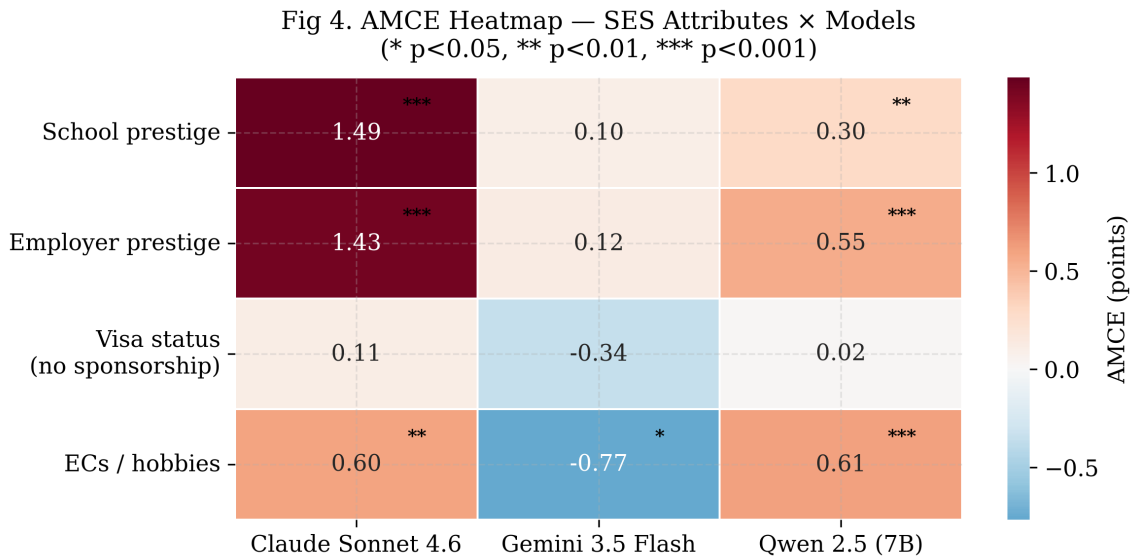


Figure 5: AMCE heatmap, Data Analyst. Red denotes advantage to the higher-SES level, blue disadvantage.

Fig 4. AMCE Heatmap — SES Attributes × Models
(* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$)

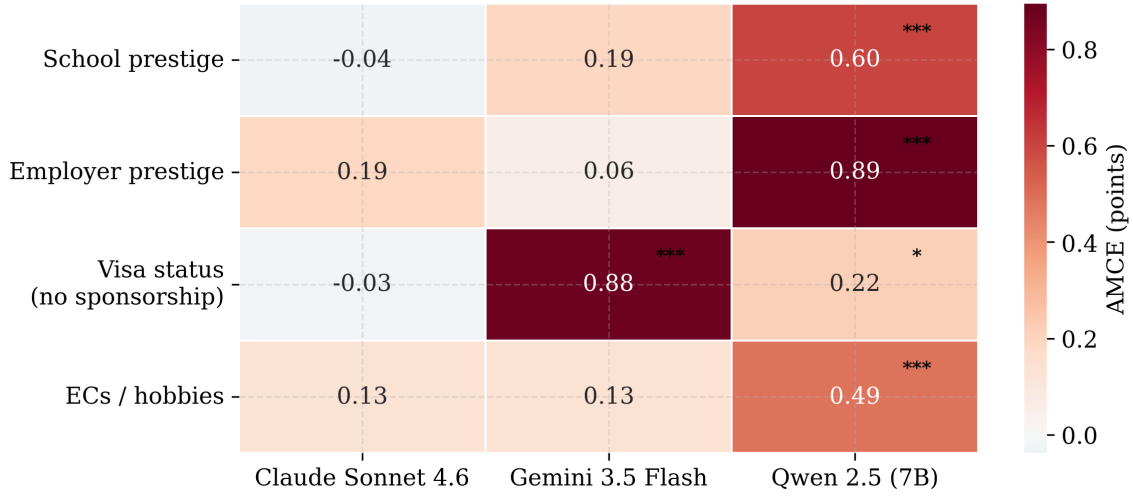


Figure 6: AMCE heatmap, Software Engineer. The concentration of Gemini 3.5 Flash’s response in the visa row, and its absence elsewhere, is visible at a glance.

Standardized magnitudes. Expressed in units of each model’s own score standard deviation, the largest effects are Qwen’s employer prestige for Software Engineers (0.236 SD), Gemini’s visa status for Software Engineers (0.227 SD), Claude’s school prestige for Data Analysts (0.184 SD), Claude’s employer prestige for Data Analysts (0.177 SD), and Qwen’s extracurriculars for Data Analysts (0.161 SD). All remaining significant effects fall below 0.16 SD. By conventional standards these are small; the shortlisting results below show why that description understates their consequences.

Composite SES association. Pearson correlations between the count of high-SES attributes (0–4) and the individual score reinforce the pattern. For Data Analysts, Claude shows $r = +0.111$ ($p < 0.001$) and Qwen $r = +0.097$ ($p < 0.001$), while Gemini shows $r = -0.018$ ($p = 0.21$), statistically indistinguishable from zero and, if anything, negative.

7.3 Bivariate Effects and Effect Sizes

The AMCE regressions in Table 5 condition on all four attributes simultaneously. Table 7 reports the corresponding bivariate contrasts — a simple comparison of high-SES against low-SES résumés for each attribute in turn — together with Cohen’s d , which places the effects on a scale independent of each model’s score dispersion. Because the design is balanced, the two sets of estimates agree closely; the value of the bivariate table is that it supplies standardised effect sizes and makes clear how small these effects are in conventional terms even where they are highly significant.

The largest standardised effect anywhere in the Data Analyst experiment is Claude’s school prestige at $d = 0.18$, which by the usual convention is below the threshold for a “small” effect.

Gemini’s significant negative extracurricular effect is $d = -0.06$; Qwen’s largest is $d = 0.16$. Section 9.4 takes up why these figures nonetheless translate into the shortlisting disparities documented in Table 10.

Table 7: Bivariate Contrasts and Standardised Effect Sizes (Data Analyst)

Attribute	Model	Mean (high)	Mean (low)	Diff.	Cohen’s d	p
School prestige	Claude Sonnet 4.6	71.54	70.09	+1.455	+0.181	< 0.001
	Gemini 3.5 Flash	82.41	82.29	+0.122	+0.010	0.730
	Qwen 2.5 (7B)	90.24	89.97	+0.274	+0.073	0.010
Employer prestige	Claude Sonnet 4.6	71.54	70.12	+1.419	+0.177	< 0.001
	Gemini 3.5 Flash	82.41	82.29	+0.117	+0.009	0.742
	Qwen 2.5 (7B)	90.39	89.84	+0.552	+0.146	< 0.001
Visa status	Claude Sonnet 4.6	70.87	70.74	+0.125	+0.016	0.584
	Gemini 3.5 Flash	82.18	82.52	-0.345	-0.028	0.329
	Qwen 2.5 (7B)	90.12	90.09	+0.024	+0.007	0.819
ECs/hobbies	Claude Sonnet 4.6	71.08	70.53	+0.551	+0.068	0.016
	Gemini 3.5 Flash	81.96	82.73	-0.769	-0.062	0.029
	Qwen 2.5 (7B)	90.41	89.81	+0.598	+0.159	< 0.001

Notes: Welch two-sample t -tests comparing the high-SES against the low-SES level of each attribute, $N = 5,000$ per model. Cohen’s d uses the pooled standard deviation. “High” for visa status denotes authorisation without sponsorship.

Table 8 summarises the relationship between the composite SES count and the score. It reinforces the central asymmetry: for Claude and Qwen, moving a résumé up the SES ladder raises its score reliably, whereas for Gemini the association is statistically indistinguishable from zero and points, if anything, in the opposite direction.

Table 8: Correlation Between Composite SES Count and Score (Data Analyst)

Model	Pearson r	r^2	p	N
Claude Sonnet 4.6	+0.111	0.0123	< 0.001	5,000
Gemini 3.5 Flash	-0.018	0.0003	0.212	5,000
Qwen 2.5 (7B)	+0.097	0.0093	< 0.001	5,000

Notes: Composite SES is the count of high-SES attributes assigned to a résumé, ranging from 0 to 4.

7.4 Shortlisting

Table 9 reports linear probability estimates.

The most consequential single effect in the study is Gemini’s visa coefficient for Software Engineers. Candidates requiring sponsorship are shortlisted 19.6% of the time against 30.1% for otherwise identical candidates who do not, a gap of 10.5 percentage points. The associated score effect, 0.88 points, is small on its face; the selection effect is not.

This gap between score effects and selection effects is a general feature of the results and follows

Table 9: Effect of SES Attributes on Shortlisting Probability (LPM, HC3 SE)

Attribute	Claude Sonnet 4.6	Gemini 3.5 Flash	Qwen 2.5 (7B)
<i>Panel A. Data Analyst</i>			
School prestige	+0.058***	+0.006	+0.026*
Employer prestige	+0.058***	+0.010	+0.050***
Visa status	+0.006	+0.000	+0.012
ECs/hobbies	+0.030**	-0.020	+0.053***
Baseline rate	0.110	0.197	0.133
<i>Panel B. Software Engineer</i>			
School prestige	-0.010	+0.008	+0.041***
Employer prestige	+0.019	+0.020	+0.063***
Visa status	+0.001	+0.105***	+0.006
ECs/hobbies	-0.006	+0.018	+0.031**

Notes: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. Coefficients are marginal effects on shortlisting probability, in probability units; +0.058 corresponds to 5.8 percentage points. Baseline rate is the model intercept, the fitted shortlisting probability for an all-low-SES candidate.

Fig 5. Shortlisting Rates by SES Attribute Level and Model

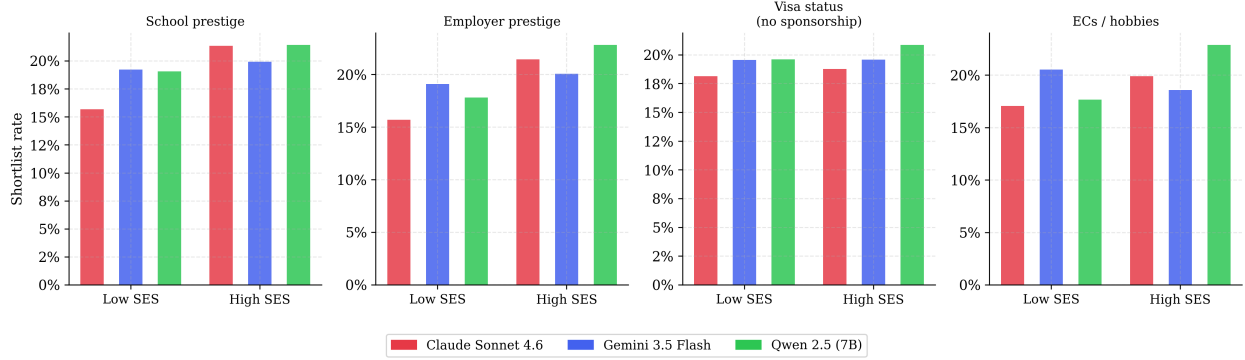


Figure 7: Shortlisting rates by attribute level and model, Data Analyst.

Fig 5. Shortlisting Rates by SES Attribute Level and Model

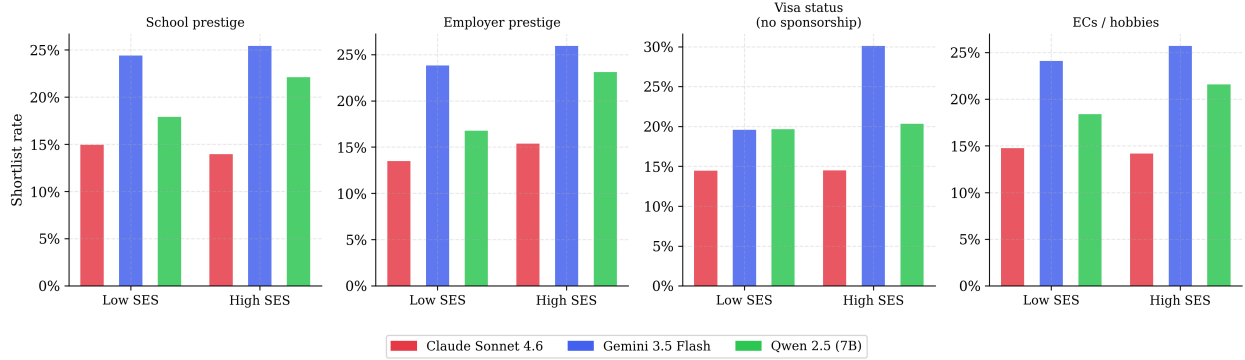


Figure 8: Shortlisting rates by attribute level and model, Software Engineer. Gemini 3.5 Flash’s visa panel shows the single largest gap observed in the study.

from the compressed score distributions. When a model’s scores are concentrated within a few points, a sub-point shift relocates a large mass of candidates across whatever internal threshold governs shortlisting. Gemini’s Software Engineer scores have $SD = 3.9$, so a 0.88-point shift is 0.23 SD applied to a dense region of the distribution. Small coefficients on compressed scales are not benign.

7.5 Adverse Impact and the Four-Fifths Rule

Table 10 applies the four-fifths benchmark to composite SES groups, comparing candidates with zero or one high-SES attribute against those with three or four.

Table 10: Shortlisting Rates and Impact Ratios by Composite SES Group

Occupation	Model	Low SES	High SES	Gap	Impact ratio	Four-fifths
Data Analyst	Claude Sonnet 4.6	15.0%	24.3%	+9.3pp	0.617	fails
	Gemini 3.5 Flash	20.5%	19.9%	−0.6pp	0.971	passes
	Qwen 2.5 (7B)	15.5%	24.3%	+8.8pp	0.638	fails
Software Engineer	Claude Sonnet 4.6	14.3%	14.4%	+0.1pp	0.993	passes
	Gemini 3.5 Flash	20.5%	29.7%	+9.2pp	0.690	fails
	Qwen 2.5 (7B)	16.3%	25.2%	+8.9pp	0.647	fails

Notes: Low SES denotes 0–1 high-SES attributes, high SES denotes 3–4. The impact ratio is the lower selection rate divided by the higher. A ratio below 0.80 is conventionally treated as evidence of adverse impact under the EEOC four-fifths rule. Applied here descriptively to a non-protected composite, not as a legal determination.

Four of the six model–occupation pairs fall below the 0.80 threshold, and the pattern across the two commercial models is a clean inversion. Claude fails decisively for Data Analysts (0.617) yet is essentially perfectly neutral for Software Engineers (0.993). Gemini 3.5 Flash is neutral for Data Analysts (0.971) and fails for Software Engineers (0.690). Qwen fails in both. The single most important practical implication of this study follows directly: an audit establishing that a model is fair for one role provides no assurance whatever about a different role in the same pipeline.

Figures 9 and 10 show the underlying score gradients across composite SES groups.

7.6 Model Heterogeneity (H5)

Table 11 reports Cochran’s Q (Section 3.4) for equality of the AMCE across the three models, separately by attribute and occupation.

Table 11: Cochran’s Q Test for Equality of AMCEs Across Models

Attribute	Data Analyst		Software Engineer	
	Q (df=2)	p	Q (df=2)	p
School prestige	23.76	< 0.001	15.33	< 0.001
Employer prestige	14.91	< 0.001	33.85	< 0.001
Visa status	1.21	0.545	32.29	< 0.001
ECs/hobbies	13.95	< 0.001	6.80	0.033

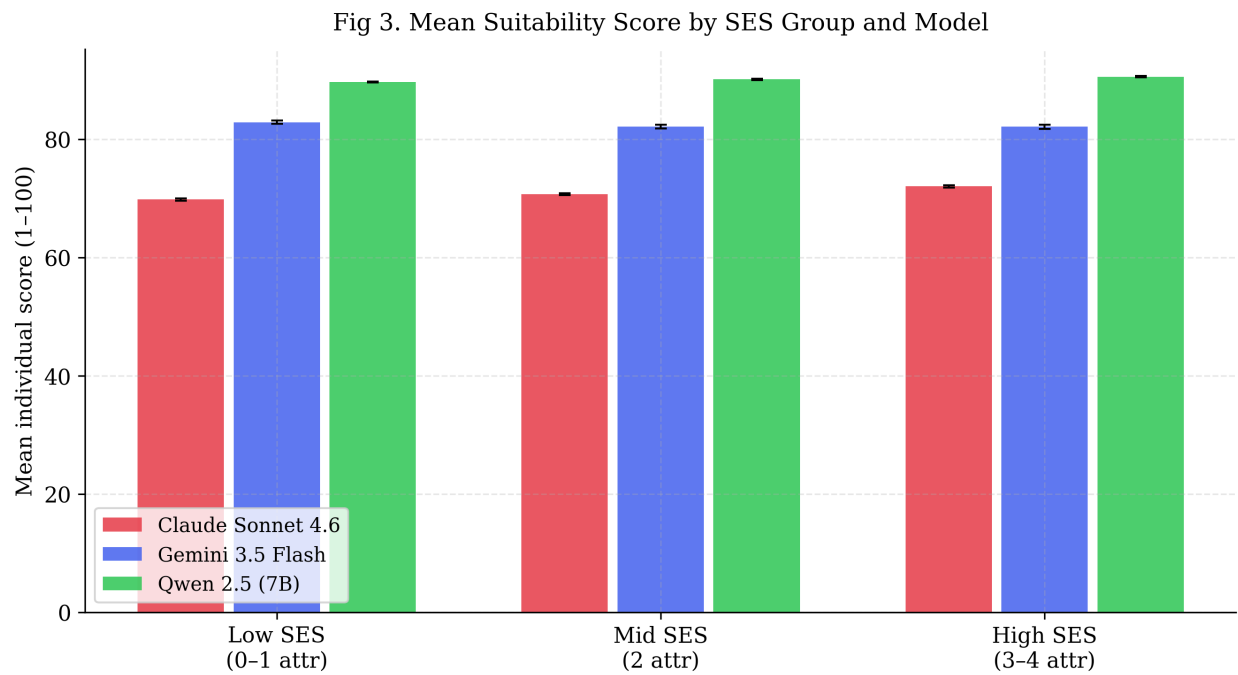


Figure 9: Mean score by composite SES group, Data Analyst. Claude's gradient is monotone and spans 2.2 points; Gemini 3.5 Flash's is slightly reversed.

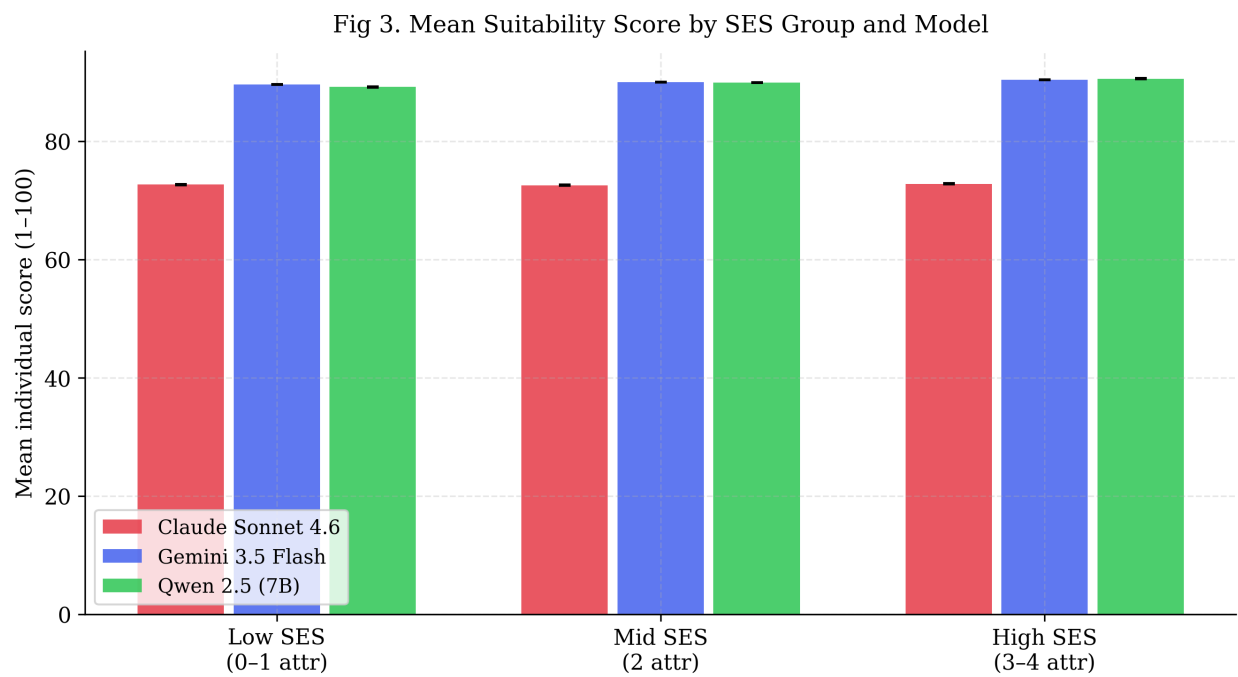


Figure 10: Mean score by composite SES group, Software Engineer. Claude is flat; Gemini 3.5 Flash and Qwen both rise.

Seven of the eight attribute–occupation combinations reject equal AMCEs across the three models at the 5% level; only visa status for Data Analysts fails to ($Q = 1.21, p = 0.545$), consistent with no model showing a significant visa effect in that occupation (Table 5). The two largest rejections are substantively the most interesting: school prestige for Data Analysts ($Q = 23.76, p < 0.001$) is driven by Claude alone, while visa status for Software Engineers ($Q = 32.29, p < 0.001$) is driven by Gemini alone. The extracurricular range spans zero rather than just magnitude—Qwen at $+0.607$ and Gemini at -0.765 assign opposite signs to the same signal—which is the qualitative disagreement, not merely a difference of degree, that the Q test at $p < 0.001$ confirms. **Verdict: H5 supported.**

Figures 11 and 12 present the joint conjoint view of score and shortlisting effects.

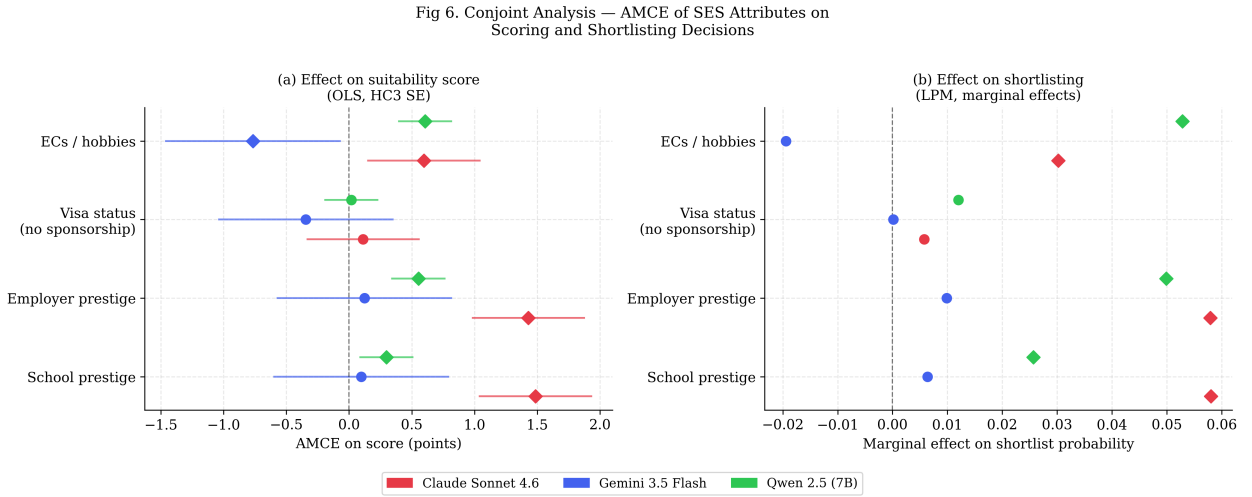


Figure 11: Conjoint view, Data Analyst: score AMCEs (left) and shortlisting marginal effects (right).

7.7 Occupational Heterogeneity (H6)

Holding the model fixed and varying only the occupation produces changes as large as those produced by varying the model, and the same Q statistic (Section 3.4) confirms it formally, now testing equality across the two occupations for a fixed model ($k = 2, df = 1$). For Claude, school prestige differs sharply across occupations ($Q = 33.25, p < 0.001$), as does employer prestige ($Q = 22.22, p < 0.001$); visa and ECs do not ($p = 0.58, p = 0.078$). For Gemini, visa status differs sharply across occupations ($Q = 10.90, p < 0.001$), as do ECs ($Q = 5.90, p = 0.015$); school and employer prestige do not ($p = 0.80, p = 0.87$). For Qwen, school and employer prestige both differ significantly across occupations ($Q = 4.11, p = 0.043; Q = 5.17, p = 0.023$), while visa and ECs do not ($p = 0.18, p = 0.42$).

Every model, in other words, has at least one attribute whose effect changes significantly with the job description, and for Claude and Qwen those attributes are exactly the ones driving their four-fifths verdicts in Table 10. Substantively: Claude moves from an impact ratio of 0.617 for

Fig 6. Conjoint Analysis — AMCE of SES Attributes on Scoring and Shortlisting Decisions

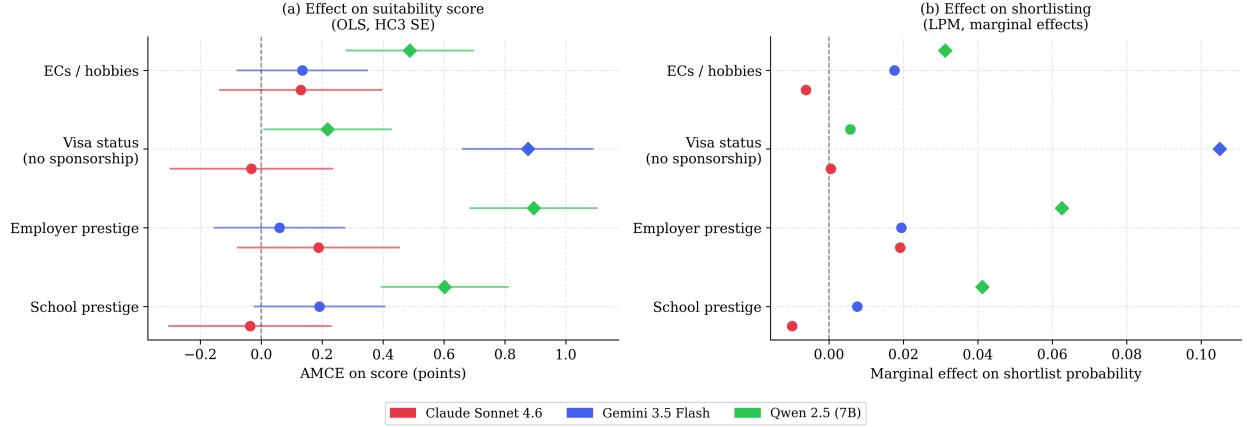


Figure 12: Conjoint view, Software Engineer: score AMCEs (left) and shortlisting marginal effects (right).

Data Analysts to 0.993 for Software Engineers; Gemini moves from 0.971 to 0.690 in the opposite direction; Qwen’s profile broadens rather than shifts, with school prestige rising from +0.298 to +0.602 and employer prestige from +0.553 to +0.894. Because the same 5,000 base résumés and an identically structured prompt are used in both occupations, with only the job description differing, the occupation itself is the operative variable. **Verdict: H6 supported.**

8 Robustness

Estimator choice. The linear probability model was adopted because batch-level shortlisting rates for Gemini and Qwen are near-constant (standard deviation below 0.003), producing quasi-complete separation under logit. LPM coefficients are directly interpretable as marginal effects [1]. Re-estimating scores with cluster-robust rather than HC3 standard errors leaves all significance conclusions unchanged.

Deduplication. The raw Data Analyst scoring file contained repeated (model, résumé) rows arising from overlapping recovery passes over the scoring response archive. Analysis deduplicates on this key, retaining the first occurrence and removing 4,762 rows to yield exactly 5,000 unique evaluations per model. Because duplication was a property of the storage pipeline rather than of the randomization, it is uncorrelated with SES assignment; but duplicated rows would nonetheless understate standard errors, so deduplication is applied before all estimation. The Software Engineer file contained no duplicates.

Design balance. Realized high-SES proportions lie within 0.02 of 0.50 for every attribute in both occupations (Table 3), so the estimates are not artifacts of unbalanced assignment.

Claude as an internal control. Claude’s Software Engineer results constitute a within-study null: identical résumés, identical prompt structure, identical estimator, and four coefficients indistinguishable from zero ($R^2 = 0.0006$; all $p > 0.16$), with an impact ratio of 0.993. This makes it implausible that the significant effects found elsewhere are artifacts of the pipeline, and it demonstrates that near-complete SES neutrality is attainable.

Excluded measures. The `pool_score` field is excluded throughout because per-résumé presentation makes it uninterpretable, and because its realized scale differs sharply across models in a manner unrelated to candidate quality. The 45-record OpenAI pilot is reported descriptively only and excluded from all regressions as far too small for stable estimation.

9 Discussion

9.1 Occupational Blind Spots

The clearest pattern in these results is that Claude and Gemini are each biased in exactly one of the two occupations I tested, and that the occupation where each one fails is different: Claude fails the four-fifths benchmark for Data Analysts and passes for Software Engineers, while Gemini does the reverse.

I don’t have access to either model’s training data, so any explanation for why is necessarily a guess, but one that fits the pattern reasonably well is that these models pick up different class heuristics for different occupations because the text describing those occupations online is itself class-coded differently. Data-analyst hiring gets discussed, in a lot of business writing, alongside consulting and finance recruiting pipelines that explicitly run through target schools; software-engineering competence, by contrast, tends to be demonstrated through concrete technical evidence—languages, systems, shipped projects—that leaves less room for a school or employer name to do independent work. Claude’s own score distribution is consistent with this story: its Data Analyst scores spread across more than 40 points, but for Software Engineers they compress into a band about half that wide ($SD = 4.8$ versus 8.1), which looks like a model applying a tighter, more rubric-like standard once the qualifications are directly checkable.

Gemini’s pattern is the mirror image, and I find it more unsettling. It is the fairest model in the study on Data Analyst prestige signals—nothing it does there clears my significance threshold—and then, on the same four signals for a different job title, it develops a large, specific reaction to visa sponsorship (Table 11 confirms this is not noise: $Q = 32.29$, $p < 0.001$). A model that looks completely clean on one audit and meaningfully biased on the next, without anyone changing anything except which job description was pasted into the prompt, is exactly the scenario that makes single-audit sign-off dangerous.

9.2 The Reverse Extracurricular Effect

Gemini’s Data Analyst coefficient on extracurriculars is the one result in this paper I did not expect going in. It is negative and significant (-0.765 , $p < 0.05$): a candidate whose résumé lists an unpaid research internship or a study-abroad program scores worse than an otherwise identical candidate who lists part-time paid work or caregiving instead. Every prior audit of class-linked résumé signals I am aware of finds the opposite direction, or no effect; I could not find a precedent for a reversal this clean.

I can think of a few explanations, and I want to be honest that I cannot distinguish between them with this design. Gemini could be treating the elite activities as weak or even negative evidence for a data-analysis role specifically—reading unpaid research as time not spent building the kind of practical, tool-based experience the job actually needs. It could reflect deliberate fine-tuning against exactly this kind of prestige signal, in which case the reversal would be evidence the intervention worked almost too well, spilling over into activities that are not really about prestige at all. Or it could be a byproduct of how compressed Gemini’s scores already are near the top of its range (skewness -2.36): with most candidates already crowded close to the ceiling, there may simply be more room for the model to push a candidate down than to push one further up. I did not build this study to adjudicate between these explanations, so I leave it as an open question rather than force an answer I cannot support.

What I can say without hedging is what it means for a candidate on the other end of that scoring pattern. The same decision—unpaid research internship instead of part-time paid work—raises the score with Claude and Qwen and lowers it with Gemini, for identical qualifications, for the same job posting. That is not a matter of degree. It tells me that whatever “class bias” means to a language model is not a fixed direction it defaults to; it is specific to the model, and apparently specific enough to flip sign on a single attribute.

9.3 Visa Status and Legal Exposure

Of everything in this paper, the visa result is the one I would flag first to anyone actually responsible for deploying one of these models. School prestige, employer prestige, and extracurricular background sit outside U.S. employment law—there is no statute a plaintiff could point to if a screener quietly favored Ivy League graduates. Visa and citizenship status is different: the Immigration and Nationality Act’s anti-discrimination provision (8 U.S.C. §1324b) and Title VII’s protection of national origin both bear on it directly.

Gemini’s only significant Software Engineer effect lands on precisely this attribute, and it is not a marginal one—a 10.5-point shortlisting gap and an impact ratio of 0.651 on that attribute alone. I want to be careful here: I am not a lawyer, and I am not claiming this rises to unlawful discrimination—sponsorship status can be a legitimate business consideration in some circumstances, and how courts would treat an LLM’s internalized version of that consideration is a genuinely open legal question, not one I can resolve in a methods paper. What I can say is that this finding sits in a different compliance category than the rest of my results, and a deployer who filed

it under “equity concern, address eventually” rather than “possible legal exposure, address now” would be misreading it.

9.4 Interpreting Small Coefficients

Every significant score effect in this paper is under a point and a half on a hundred-point scale, and no model’s R^2 climbs above 0.026. I do not think that is a weakness of the design so much as a predictable consequence of it: I am holding four SES attributes against everything else a résumé could plausibly signal, deliberately, so that whatever a coefficient does capture is class response and nothing else. A small coefficient here is not a small effect diluted by noise; it is a pure, isolated one.

I would also push back on using R^2 as the yardstick at all. Hiring is a threshold decision, not a continuous outcome—nobody gets half an interview. The number that matters is how many candidates a signal moves across whatever line separates a callback from a rejection, and by that measure four of my six model–occupation pairs cross the regulatory line regulators actually use. Claude’s Data Analyst impact ratio of 0.617 does not come from some outlier coefficient; it comes from effects that average about a point and a fifth.

9.5 Implications for Deployment

Three things follow from this, and I think they are fairly concrete. First, picking a model for a hiring pipeline is already an equity decision, whether or not anyone frames it that way—the models here differ by more than a full point on identical inputs and, on extracurriculars, do not even agree on the sign. Second, auditing has to happen per role, not per model: every commercial model in this study passed at least one audit and failed another, so a green light on one job posting says nothing about the next one. Third, whatever gets reported should be a selection-stage number, not a score-stage one—I have shown that a coefficient can look negligible in points and still move a fifth of the applicant pool across the shortlist line once the score distribution is tight enough.

Limitations

A few limitations bound how far these results travel. The résumés are synthetic, which buys the experimental control the design needs at some real cost to realism. One of the three audited models also generated the résumés; I hard-coded the SES attributes myself rather than letting the model choose them, which limits but does not eliminate the concern, and replicating with an independent generator is the most obvious next step. Because résumés were scored one at a time, **shortlisted** reflects each model’s own internal cutoff rather than a genuine constrained top- k selection, so I can compare SES levels within a model but not shortlist rates across models at face value. I used one prompt per occupation and did not test sensitivity to prompt wording. I covered two occupations and one national labor market. And I have no human baseline—I cannot say whether these models

are more or less class-biased than the recruiters they would be replacing, which is, in the end, the comparison that would actually matter for a hiring decision.

10 Conclusion

This paper set out to audit three language models for class bias in résumé screening across two occupations, using a randomized conjoint design and 30,045 scored evaluations with qualifications, race, and gender held fixed. The bias I found was not a single, fixed property of any model—it depended jointly on which model was doing the screening and which job it was screening for.

Claude rewards school and employer prestige for Data Analysts, lifting shortlisting from 15.0% to 24.3% and landing an impact ratio of 0.617, but is statistically indistinguishable from neutral for Software Engineers. Gemini shows the opposite pattern: 0.971 for Data Analysts, then a 10.5-point shortlisting gap concentrated on visa sponsorship for Software Engineers. Qwen fails the four-fifths benchmark in both. And Gemini’s one significant Data Analyst effect runs backward from what either other model does, penalizing rather than rewarding elite extracurriculars—direct sign disagreement on the same signal between models trained by different labs.

The attribute-level hypotheses, H1 through H4, were therefore only ever partially supported: true for some models and jobs, false for others, and in Gemini’s case, true in reverse (Table 6). The heterogeneity hypotheses, H5 and H6, held up strongly and consistently, each confirmed by a formal test rather than argued from magnitude alone. If there is one practical conclusion worth taking away, it is a narrow one: no model here was safe across the board, every commercial model had at least one occupation where it failed, and a fairness result from one job posting does not transfer to the next one by anything but coincidence. Auditing needs to happen per model and per role, and it needs to be reported at the stage where the consequences land—the shortlist, not the score.

Acknowledgments

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